

DEMONSTRATION AND ANALYSIS OF LLM HALLUCINATIONS

Question 7 – Implementation

1. Question

Demonstrate and analyze hallucinations in a locally running Large Language Model using suitable prompts and reference information.

2. Aim

To demonstrate how hallucinations can be identified in a locally running Large Language Model by comparing its generated answers with trusted reference information and testing questions for which the reference does not contain an answer.

3. Technologies Used

- Python 3.12.10
- Ollama 0.33.2
- Llama 3.2 – local Large Language Model
- Ollama Python library
- VS Code

4. Project Structure

Local_LLM_Text_Generation/

- |— venv/ – Python virtual environment
- |— app.py – Question 1 text generation
- |— summarization.py – Question 2 text summarization
- |— qa.py – Question 3 question answering
- |— translation_paraphrasing.py – Question 4 translation and paraphrasing
- |— ollama_text_generation.py – Question 5 Ollama text generation
- |— ollama_qa.py – Question 6 Ollama question answering
- |— hallucination_demo.py – Question 7 hallucination analysis

5. Implementation Steps

1. Use the existing project folder and virtual environment.
2. Ensure Ollama and the local Llama 3.2 model are available.
3. Create a reference-information section containing known facts.
4. Create prompts that instruct the model to use only the supplied reference.
5. Test a question whose answer is explicitly present in the reference.
6. Test questions whose answers are not present in the reference.
7. Instruct the model to state that information is unavailable when the reference does not contain the answer.
8. Compare the generated answers with the reference information.

9. Identify any unsupported or fabricated claims as potential hallucinations.
10. Display the test results and analysis in the terminal.

6. Python Implementation

```
import ollama

def ask_llm(prompt):
    response = ollama.chat(
        model="llama3.2",
        messages=[
            {
                "role": "user",
                "content": prompt
            }
        ]
    )

    return response["message"]["content"]

print("=" * 70)
print("LLM HALLUCINATION DEMONSTRATION USING OLLAMA")
print("=" * 70)

reference = """
The Smart Campus project was developed by the Computer Science
Department in 2025. The project uses IoT sensors to monitor
classroom temperature and energy consumption. The system sends
sensor readings to a local dashboard for monitoring.
"""

print("\nREFERENCE INFORMATION")
print("-" * 70)
print(reference)

print("\n" + "=" * 70)
print("TEST 1: QUESTION WITH REFERENCE INFORMATION")
print("=" * 70)

prompt1 = f"""
Answer the question using only the reference information below.

Reference Information:
{reference}

Question:
What does the Smart Campus project monitor?

If the answer is not present in the reference, say:
"Information not available in the reference."

```

```
"""
```

```
answer1 = ask_llm(prompt1)
```

```
print("\nAnswer:")
```

```
print("-" * 70)
```

```
print(answer1)
```

```
print("\n" + "=" * 70)
```

```
print("TEST 2: UNKNOWN INFORMATION")
```

```
print("=" * 70)
```

```
prompt2 = f"""
```

```
Use only the reference information below.
```

```
Reference Information:
```

```
{reference}
```

```
Question:
```

```
Who was the project manager of the Smart Campus project?
```

```
If the answer is not present in the reference, say:
```

```
"Information not available in the reference."
```

```
"""
```

```
answer2 = ask_llm(prompt2)
```

```
print("\nAnswer:")
```

```
print("-" * 70)
```

```
print(answer2)
```

```
print("\n" + "=" * 70)
```

```
print("TEST 3: HALLUCINATION-PRONE QUESTION")
```

```
print("=" * 70)
```

```
prompt3 = f"""
```

```
Answer the following question.
```

```
Do not assume facts that are not provided.
```

```
Reference Information:
```

```
{reference}
```

```
Question:
```

```
How many students developed the Smart Campus project?
```

```
If the information is not available, clearly state:
```

```
"Information not available in the reference."
```

```
"""
```

```
answer3 = ask_llm(prompt3)
```

```
print("\nAnswer:")
```

```
print("-" * 70)
```

```
print(answer3)
```

```
print("\n" + "=" * 70)
print("HALLUCINATION ANALYSIS")
print("=" * 70)
```

```
print("""
Test 1:
The answer can be verified directly from the reference information.
```

```
Test 2:
The project manager is not mentioned in the reference.
A correct responsible response should state that the information
is unavailable rather than inventing a person's name.
```

```
Test 3:
The number of students is not provided in the reference.
Any specific number generated by the model would be unsupported
information and would represent a hallucination.
```

```
Therefore, comparing the generated answers with the reference
information helps identify unsupported or fabricated claims.
""")
```

7. Working Principle

The system provides the local LLM with reference information and carefully designed prompts. Test 1 asks for information explicitly contained in the reference. Tests 2 and 3 ask for information that is absent. The generated answers are then compared with the reference to determine whether the model remains grounded or produces unsupported information.

8. Test Cases and Analysis

Test	Reference Contains Answer?	Observed Output	Analysis
Test 1	Yes	Classroom temperature and energy consumption.	Supported by the reference; no hallucination observed.
Test 2	No	Information not available in the reference.	Responsible response; no unsupported person name was invented.
Test 3	No	Information not available in the reference.	Responsible response; no unsupported student count was invented.

9. Reference Information

The Smart Campus project was developed by the Computer Science Department in 2025. It uses IoT sensors to monitor classroom temperature and energy consumption, and sends sensor readings to a local dashboard.

10. Input

Test 1: What does the Smart Campus project monitor?

Test 2: Who was the project manager of the Smart Campus project?

Test 3: How many students developed the Smart Campus project?

11. Output

Test 1 returned the supported information that the project monitors classroom temperature and energy consumption. Tests 2 and 3 returned “Information not available in the reference,” correctly avoiding unsupported details.

12. Hallucination Analysis

A hallucination occurs when a language model generates information that is unsupported by the available evidence or reference. In this demonstration, the model was instructed to rely only on the supplied reference. All three observed responses remained consistent with the reference, so no hallucination was observed in the demonstrated run.

13. Result

Thus, hallucination behavior in a locally running Llama 3.2 model was successfully demonstrated and analyzed using reference information and suitable prompts. The experiment showed that grounding the model with reference information and explicitly requiring it to acknowledge missing information can help reduce unsupported responses.

14. Conclusion

The experiment demonstrates the importance of reference information and responsible prompting when using local Large Language Models. Comparing generated answers with trusted information provides a practical method for detecting unsupported claims and evaluating model reliability.